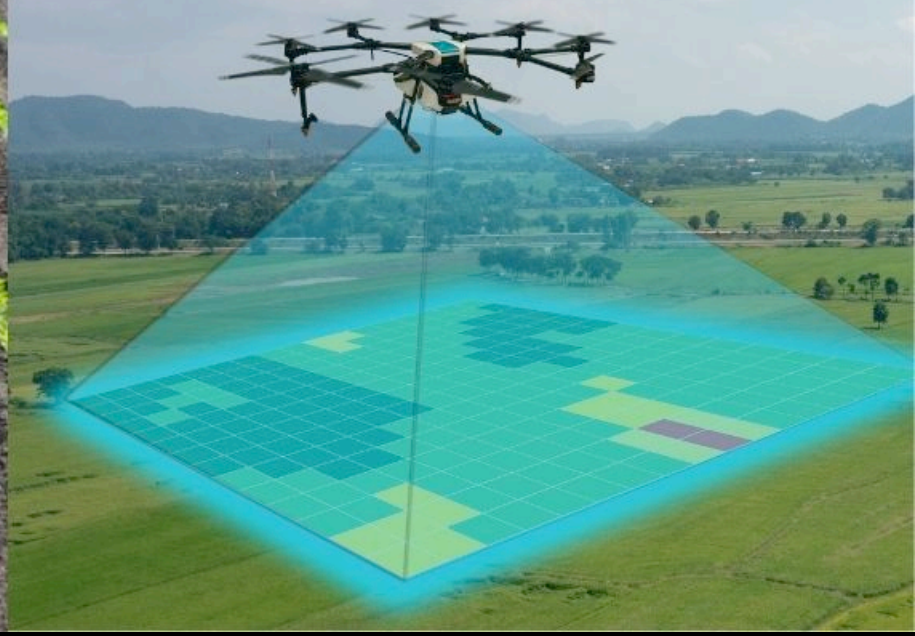




Holos

Improve Farming Profits through Sustainable Insights based on Crop Science and Machine Learning

www.gaiadhi.com

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About GaiaDhi

- Early stage, self-funded US-S Corp AgTech startup
- Comprising of computer professionals
- **Engaged with TNAU for agronomy expertise (since 2020)**
- Released **Holos cMaaS** (crop Modeling as a Service) platform and associated products

Team

Karthik K

Dr. Shankar S

Vasu V

T Ashok (India)

Abhinav

Experience

Multiple years of R&D of system-level products

Deep expertise in SW & HW

Publications, patents

Startup & corporate (NASA, Apple, Vicarious, Intuit, HP) experience

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Crop Simulation Models and Holos' Benefits

CSMs : software that simulates the development of crops taking into consideration interactions between {G, E, M}. Used to forecast yield, nutrient and water consumption, trends across varieties, etc., as a precursor to cultivation.

Analogy: software models that are used to simulate crash tests in the automotive industry, to choose materials/design, in lieu of crash testing physical prototypes, thus reducing time and money.

<u>CSM (eg., DSSAT) Limitations</u>

- Point based
- Weather, soil data to be generated or fed manually
- Manual interpretation (or custom code) of voluminous output
- Manual effort for custom charting
- Software Installation/Maintenance
- Complex User Interface
- Not aware of in-field situations

<u>Holos Benefits: Bring the power of Crop Science to the Agri ecosystem</u>

- Spatio-Temporal Solution: area agnostic
- Integrated with on-line weather and soil services/repos (bias-corrected, location specific, future**)
- Making DSSAT in-field aware through remote sensing*
- Execute historical simulations for calibration and comprehensive analysis
- Multi-disciplinary analysis based on objective functions, to aid decision making*
- Custom charting*
- Cloud service - no on-premise maintenance
- Simplified user interface, with email responses

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* first version released, ** GCM models

Goals & Market Segments

Accomplished

- ★ Software Platform to Expedite Research, through automation and cloud services
- ★ Pre Season Sustainable Recommendations (Advisory)
- ★ Yield forecast/trend per block for policy making (Gov)

Mid term

Timely In-Season Recommendations/Actions using in-field monitoring (Farming Org/Co-Op/ Individual farmer)

Long term

- Demand based Insights (Individual farmer)
- Supply Chain Insights (Logistics)

Yield validated through Kuruvai and Samba seasons through TNAU

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Use Cases

for Seed Companies

- Better sourcing (Rasi Seeds)
- Better pre-season recommendations to farmers

for Crop Breeders

Augment METs through simulation (CCRRF)

for Fertilizer Companies

Augment test labs across multi regions over life cycle of crop

for Gov Agencies: Yield forecast (Fasal)

for Farmers

Sustainable recommendations to meet target yield/stability

for Logistics Companies

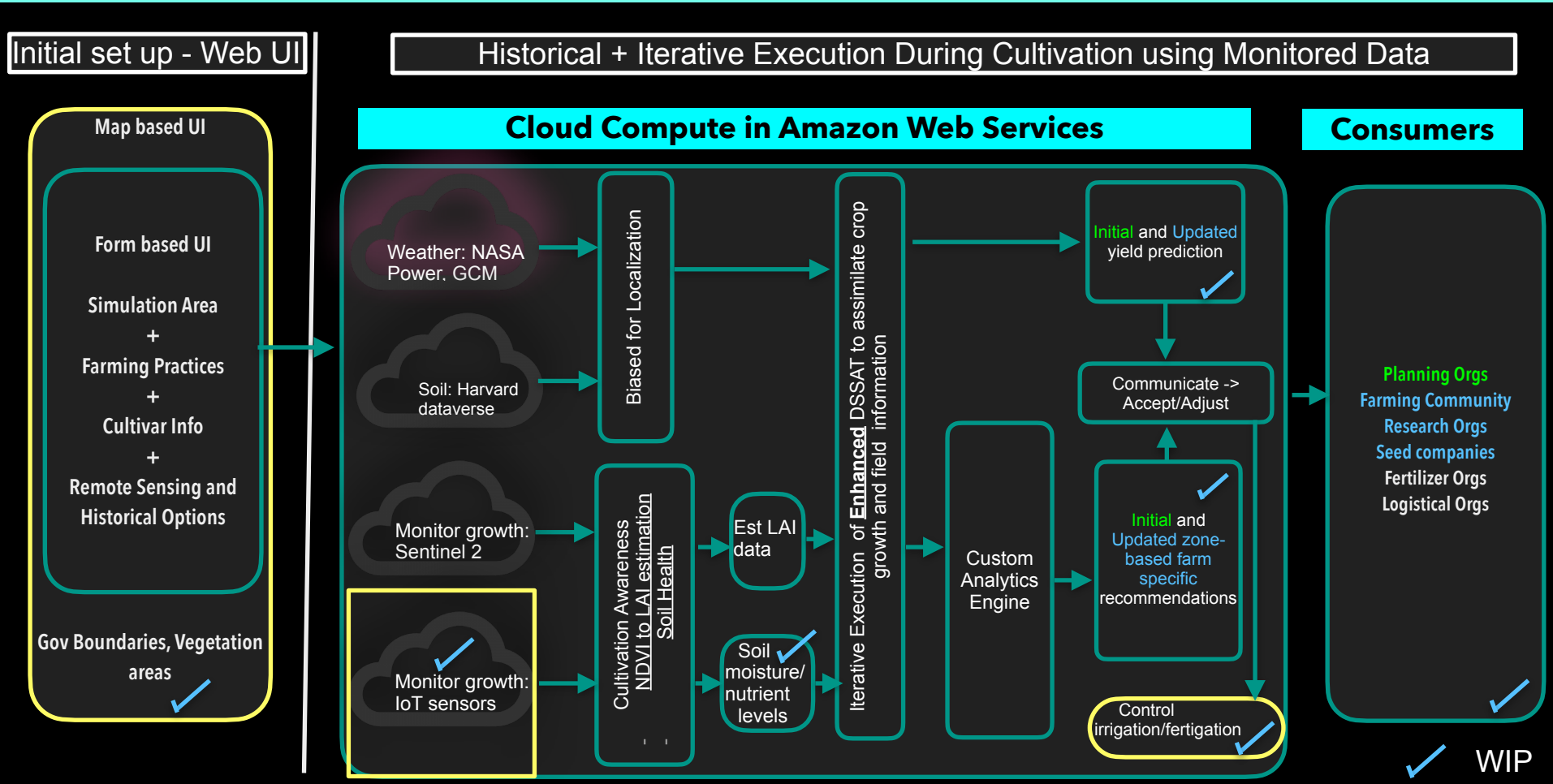
- Optimize storage locations
- Route transportation lines

for Research Institutions

- Remote Sensed Crop Growth Ingestion into DSSAT (USQ)
- Validation of Regenerative Practices for Climate Smart Cotton (TAMU)

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Holos System Architecture



Sample Cross Domain Pre-Season Recommendation

Optimal Rice Variety and Planting Date to Maximize Yield

Question: For my paddy farm, what is an optimal variety and planting date to maximize my yield based on environmental constraints

Holos Simulation Results

Simulation over 10 Year(s) without LAI assimilation for target planting date of 2023 - 4 -14.

Attached charts:
Yield performance of Cultivars Averaged over 10 Years
Holos Recommendations

We tried to identify the value of Cultivar in order to maximize Unadjusted Yield (kg/ha)) while satisfying the following constraints

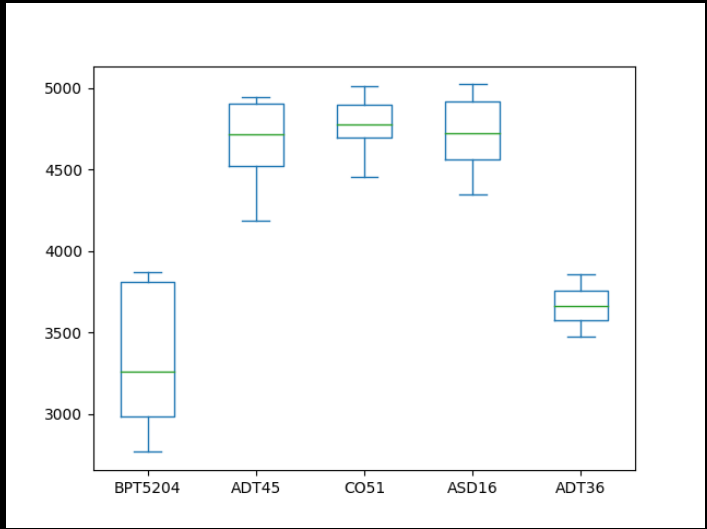
- Yield Day should be less-than 126
- CO2 Emitted (kg/ha) should be less-than 1500
- Nitrogen Used (kg/ha) should be less-than 100

One solution was found where Cultivar = ASD16
Optimal (maximize)) value of Unadjusted Yield (kg/ha)) = 4735.28

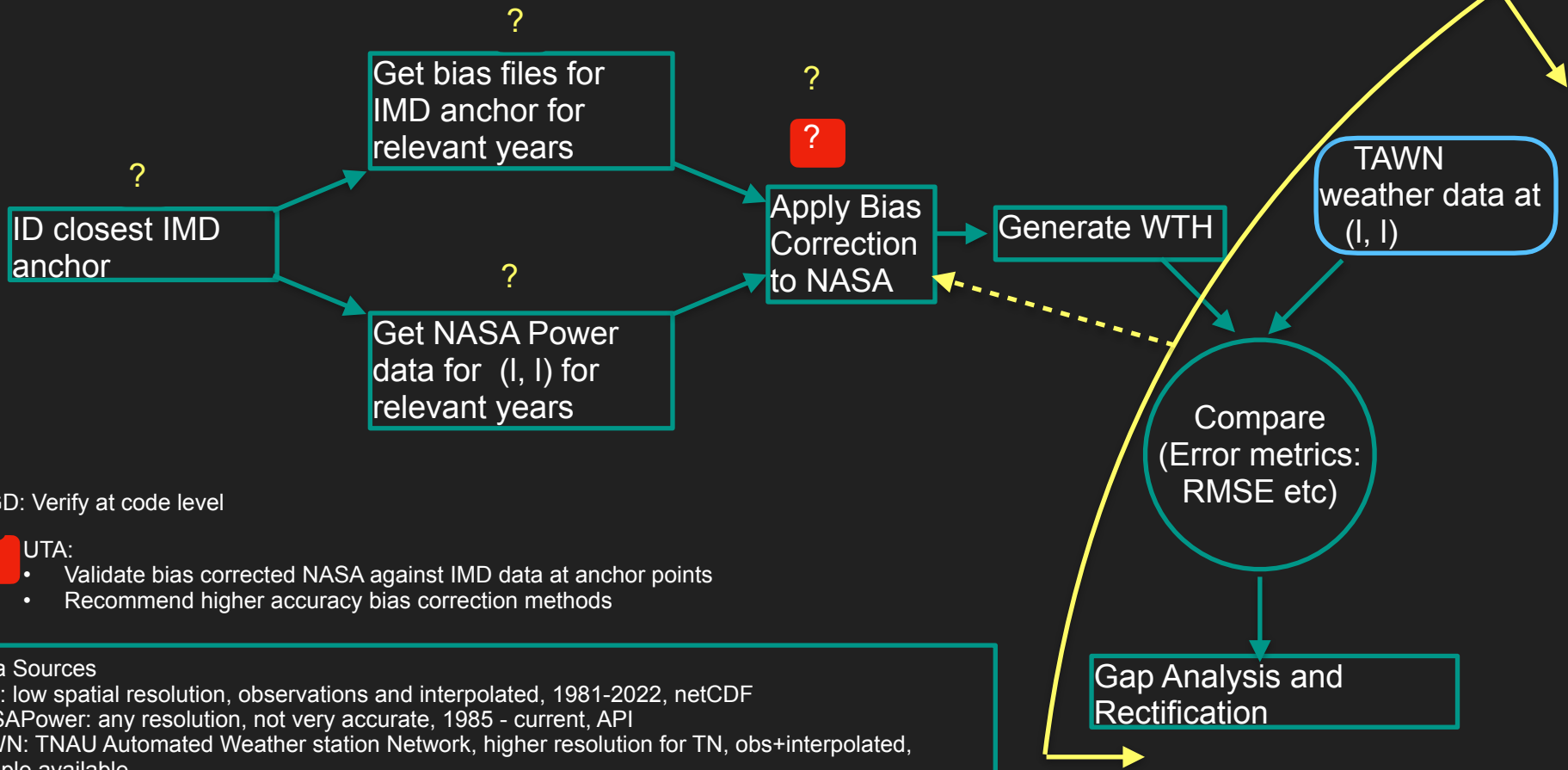
We tried to identify the value of Planting Date in order to maximize Unadjusted Yield (kg/ha)) while satisfying the following constraints

- Cultivar should be equal ASD16

One solution was found where optimal Planting Date = 21/04/2023
Optimal (maximize)) value of Unadjusted Yield (kg/ha)) = 4282.68



Holos NASA Weather Pipeline for (lat, long) (l, l) - Validate and Harden



? GD: Verify at code level

? UTA:

- Validate bias corrected NASA against IMD data at anchor points
- Recommend higher accuracy bias correction methods

Data Sources
IMD: low spatial resolution, observations and interpolated, 1981-2022, netCDF
NASAPower: any resolution, not very accurate, 1985 - current, API
TAWN: TNAU Automated Weather station Network, higher resolution for TN, obs+interpolated, sample available